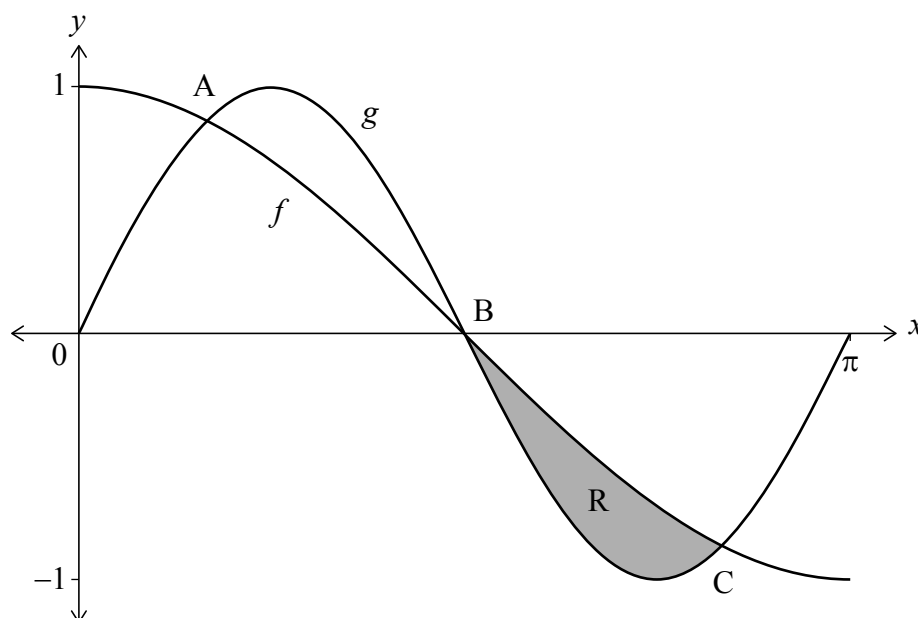


4. [Maximum mark: 7]

Consider the functions $f(x) = \cos x$ and $g(x) = \sin 2x$, where $0 \leq x \leq \pi$.

The graph of f intersects the graph of g at the point A, the point B $\left(\frac{\pi}{2}, 0\right)$ and the point C as shown on the following diagram.



- (a) Find the x -coordinate of point A and the x -coordinate of point C. [3]

The shaded region R is enclosed by the graph of f and the graph of g between the points B and C.

- (b) Find the area of R. [4]

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